

GRADE 11 PHYSICS DPP-1

Q.1 NEET UG 2013 6. Three blocks with masses m , $2m$ and $3m$ are connected by strings, as shown in the figure. After an upward force F is applied on block m , the masses move upward at constant speed v . What is the net force on the block of mass $2m$? (g is the acceleration due to gravity) m $2m$ $3m$ v F (1) $6mg$ (2) zero (3) $2mg$ (4) $3mg$

Q.2 Two bodies of masses 4 kg and 5 kg are acted upon by the same force. If the acceleration of lighter body is 2 m/s^2 , then the acceleration of the heavier body is :-(1) 4.2 m/s^2 (2) 3.6 m/s^2 (3) 2.4 m/s^2 (4) 1.6 m/s^2

Q.3 An object with a mass 10 kg moves at a constant velocity of 10 m/s . A constant force then acts for 4 seconds on the object giving it a speed of 2 m/s in opposite direction. The acceleration produced is :-(1) 3 m/s^2 (2) -3 m/s^2 (3) 0.3 m/s^2 (4) -0.3 m/s^2

Q.4 A ship of mass $3 \times 10^7\text{ kg}$ initially at rest is pulled by a force of $5 \times 10^4\text{ N}$ through a distance of 3 m . Neglecting friction, the speed of the ship at this moment is :-(1) 3.0 m/s (2) 1.5 m/s (3) 0.1 m/s (4) 2 m/s

Q.5 In Newton's second Law $\vec{F} = m\vec{a}$ (for constant mass m), $\vec{a}$ is the acceleration of the mass m with respect to (1) any observer (2) any inertial observer (3) an observer at rest only (4) an observer moving with uniform speed only

Q.6 A balloon of mass M is descending with a constant acceleration $g/3$. When a mass m is released from the balloon it starts rising with the same acceleration $g/3$. The value of m is (Assuming that its volume does not change) :-(1) $M/2$ (2) $M/4$ (3) $4M$ (4) $2M$

Q.7 Gravel is dropped onto a conveyer belt at a rate of 0.5 kg/s . The extra force required in newton to keep the belt moving at 2 m/s is :-(1) 1 N (2) 2 N (3) 4 N (4) 0.5 N

Q.8 A block of metal weighing 2 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 kg/s with a speed of 5 m/s . The initial acceleration of the block will be :-(1) 2.5 m/s^2 (2) 5 m/s^2 (3) 10 m/s^2 (4) 15 m/s^2